AGRICULTURAL ROBOTS: The Next-Gen Farmers

Agriculture is the base of our survival, the source of what we consume. Traditionally, agriculture has been a 'manned' sector. But of late challenges in the form of labour shortage threaten its very survival. So, should agriculture finally give up its stubbornness and adopt technology? Let us find out

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Agricultural robots are semi-autonomous machines capable of carrying out soil preparation, seeding, weeding, pest control, and other agricultural tasks. By automating many of the labour-intensive, repetitive tasks, farming efficiency can be increased.

In recent times, agriculture has been facing immense challenges. While the global population continues to grow, available agricultural land has been declining. The Covid-19 crisis has further aggravated the issue. Labour shortage had always been a major problem, but with the recent spate of lockdowns and labour movement restrictions, things have become worse. Adding insult to injury are the skyrocketing labour prices.

> As per UN projections, the world population can reach 8.5 billion by 2030, which can further increase to 9 billion by 2050. To adequately feed these many people, global food production has to increase by 70%. So, anything that can help boost agricultural productivity for improved food security will be a welcoming respite.

Much like in other major industries, technology offers a solution. The good news is that robotics and automation costs have been falling throughout the past two decades, which makes the possibility of automation too good to ignore.

Existing trends in agricultural robotics

As of autumn 2021, nearly fifty field and harvest robots were commercially avail-

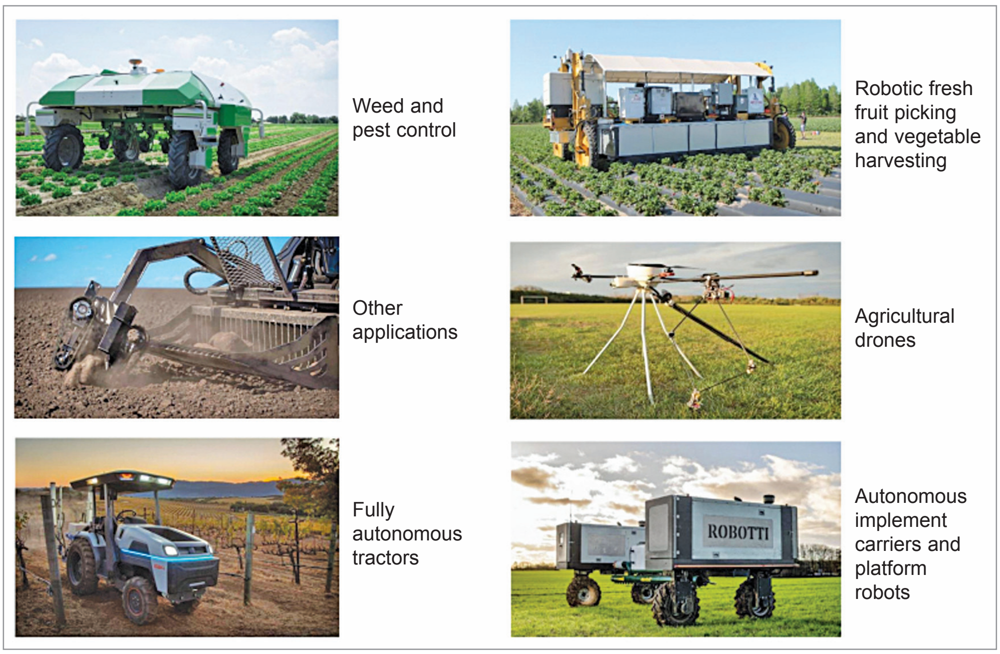

Fig. 1: Common types of agricultural robots (Credit: IDTechEx)